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UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER ONE EXAMINATIONS, 2024/2025

TIME: 2 HOURS

COURSE CODE: TEUP 305

COURSE TITLE: INTEGRATED SCIENCE III FOR UPPER PRIMARY

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and three from Section B.

SECTION A
[25 marks]
Answer all the questions in this section.

1. What process allows water vapor to become liquid water?
   a) Evaporation
   b) Condensation
   c) Sublimation
   d) Freezing

2. Which organ system is responsible for oxygen distribution?
   a) Circulatory system
   b) Digestive system
   c) Respiratory system
   d) Nervous system

3. What is the primary function of flower petals?
   a) Attract pollinators
   b) Produce seeds
   c) Store nutrients
   d) Support stems

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4. Which component makes up approximately 78% of air?
   a) Nitrogen
   b) Oxygen
   c) Carbon dioxide
   d) Argon

5. What type of material allows partial light transmission?
   a) Opaque
   b) Reflective
   c) Transparent
   d) Translucent

6. How do plants exchange gases through their leaves?
   a) Through stomata
   b) Through roots
   c) Through stems
   d) Through flowers

7. What is the function of red blood cells?
   a) Transport oxygen
   b) Fight infection
   c) Form clots
   d) Produce hormones

8. During which process does a solid change directly to gas?
   a) Sublimation
   b) Melting
   c) Condensation
   d) Evaporation

9. What causes a rainbow to form in the sky?
   a) Cloud reflection
   b) Light refraction in water droplets
   c) Atmospheric pressure
   d) Wind patterns

10. Which teaching method is most effective for science inquiry?
    a) Experimental method
    b) Lecture method
    c) Reading method
    d) Writing method

11. What is the primary role of fruit in plants?
    a) Attract pollinators
    b) Store water
    c) Protect and disperse seeds
    d) Produce energy

Student's Index No.: _____ Signature: _____

12. What determines the color of objects we see?
    a) Reflected wavelengths
    b) Object temperature
    c) Viewing angle
    d) Light intensity

13. Which of the following is NOT a factor affecting evaporation?
    a) Temperature
    b) Wind speed
    c) Surface area
    d) Air pressure

14. What is the main function of the ozone layer?
    a) UV protection
    b) Temperature control
    c) Rain formation
    d) Wind generation

15. How are seeds typically dispersed by animals?
    a) Explosive pods
    b) Fleshy fruits
    c) Wing-like structures
    d) Hook-like structures

16. Which co-teaching model involves divided instruction?
    a) Parallel teaching
    b) Solo teaching
    c) Team teaching
    d) Alternative teaching

17. What process occurs during plant fertilization?
    a) Gamete fusion
    b) Pollen formation
    c) Seed dispersal
    d) Root growth

18. Which state of matter has no fixed shape or volume?
    a) Plasma
    b) Liquid
    c) Solid
    d) Gas

19. Which organ system controls body movement?
    a) Muscular system
    b) Digestive system
    c) Respiratory system
    d) Endocrine system

Student's Index No.: _____ Signature: _____

20. Which of the following gases is required for photosynthesis?
    a) Carbon dioxide
    b) Nitrogen
    c) Hydrogen
    d) Argon

Part 2: Fill-in Questions

21. The primary source of energy in the universe is the …
22. Digestion of carbohydrate start in the …
23. The tiny pores on leaf surfaces that facilitate gaseous exchange are called …
24. The process whereby solids change into the liquid state is termed …
25. The enzyme which converts starch into maltose is …

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SECTION B

[75 marks]

Answer <u>three</u> questions from this section.

1. a. State 3 uses of flowers [6 marks]
   b. Which agents are responsible for the dispersal of the following?
      α) Guava
      β) Coconut
      γ) Seeds of cotton [6 marks]
   c. Compare and contrast the different states of matter, explaining how energy relates to state changes. Use specific examples to illustrate each transition. [12 marks]
   d. State the function of the stamen of a flower. [1 mark]

2. a. Describe the water cycle and its importance to life on Earth. Include the role of state changes and explain how human activities impact this cycle. [12 marks]
   b. Discuss three importance of practical demonstrations in teaching science concepts. [9 marks]
   c. Define pollination. [2 marks]
   d. Explain sublimation. [2 marks]

3. a. Explain each of the following terms:
      α) Translucent object
      β) Opaque object
      γ) Transparent objects [6 marks]
   b. Give an example of a translucent object [1 mark]
   c. Discuss any three things you will do to ensure water conservation in your community. [6 marks]
   d. α) Mention four parts of the flower. [4 marks]
      β) State the function of the part mentioned in d α) above [8 marks]

4. a. Discuss three advantages of dispersal of seeds and fruits. [6 marks]
   b. Differentiate between cross pollination and self-pollination. [6 marks]
   c. Give three characteristics of insect pollinated plants. [6 marks]
   d. Give three characteristics of wind pollination plants. [6 marks]
   e. Name an example of solid that sublimes [1 mark]

5. a. i. Discuss any two learning styles of early adolescents. [4 marks]
      ii. State four importance of co-teaching. [8 marks]
   b. Discuss three advantages of dispersal of seeds and fruits. [6 marks]
   c. Explain three environmental factors that affect plant growth and development. [6 marks]
   d. Name an example of a flowering plant. [1 mark]

END OF PAPER